

MOHAMMED EMAD NEGM

Software Engineer | React · TypeScript · 3D Web · AI Systems

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PROFESSIONAL SUMMARY

Software engineer and third-year Software Engineering student who ships production web applications end to end — a 3D storefront benchmarked at 2,650 req/s (p99 ~46 ms), a real-time acoustic mapper with sub-50 ms processing, and a multi-tenant ERP. Strong in React, TypeScript, WebGL and REST APIs, with a bias toward mobile-first UI and measured performance.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, C++17, Dart, HTML5, CSS3

Front-End: React, Vite, Tailwind CSS, Zustand, Responsive & Mobile-First UI

3D & Animation: React Three Fiber, Drei, Three.js, GSAP, ScrollTrigger, Framer Motion, Lenis

Back-End & Data: Node.js, Express, FastAPI, REST APIs, MongoDB, SQLite, Firebase

AI & Real-Time: LangGraph, Multi-Agent Orchestration, AI Model Routing, WebSockets

Testing & Tooling: Vitest, Git, GitHub, Docker, CMake, Capacitor, Flutter, Figma

SELECTED PROJECTS

VÉRA Fragrances — Luxury 3D E-Commerce Platform

Live: vera-fragrances.vercel.app | Code: github.com/negm-7-4/vera-fragrances

- Shipped a mobile-first storefront with a scroll-driven 3D bottle film, 13-product catalogue, cart, cash-on-delivery checkout and an owner dashboard for products, orders and analytics.
- Load-tested the API at 2,650 requests/second across 50 concurrent users: 0 errors, p99 latency ~46 ms.
- Capped rendering at three WebGL contexts with device-tier quality scaling, keeping low-end phones smooth.
- Tech: React, TypeScript, React Three Fiber, GSAP, Framer Motion, Node.js, Express, SQLite, Vitest, Docker

SAMS — Multi-Agent Virtual AI Office

Code: github.com/negm-7-4

- Orchestrated three model families as one delivery team (Claude implements, GPT plans, Gemini tests) that exchange context, review each other's output and resolve blockers autonomously.
- Built the virtual office around them: workstations, meeting rooms, a shared whiteboard and real-time collaboration over WebSockets.
- Tech: React, Three.js, FastAPI, LangGraph, WebSockets, Multi-Model AI Routing

Acoustic Room Mapper — Real-Time Acoustic Geometry Detection

Code: github.com/negm-7-4/Acoustic-Room-Mapper

- Generates live 2D room floor plans from acoustic responses with sub-50 ms processing latency.
- Bridged a Flutter Material 3 client to a Python Flask API and C++17 DSP engine (FFT, cross-correlation); four visualisation modes, Arabic RTL and JSON export.
- Tech: Flutter, Dart, Python, Flask, C++17, CMake, FFT, REST APIs

Nexora ERP — Multi-Tenant Business Management SaaS

Code: github.com/negm-7-4/nexora-erp

- Built an ERP and SaaS platform spanning sales, inventory, manufacturing, HR, CRM, finance reporting and AI-assisted actions.
- Implemented tenant isolation, role-based permissions, JWT auth, offline-first storage with cloud sync and Android packaging via Capacitor.
- Tech: React, Vite, Node.js, Express, MongoDB, Firebase, IndexedDB, JWT, Docker, Capacitor

EDUCATION

B.Sc. in Computers & Information — Software Engineering

Al-Ryada University for Science and Technology (RST), Sadat City | Third year in progress | CGPA 3.39 / 4.0

CERTIFICATIONS & LANGUAGES

Meta (Coursera): Introduction to Front-End Development; Programming with JavaScript; Version Control

Languages: Arabic (Native), English (B1 Intermediate)